

CSUC201 Fundamentals of Data Structures and Algorithms

25DIT023



CHARUSAT[®]
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LAB MANUAL

CSUC201 Fundamentals of Data Structures and Algorithms



Practical-4.1**Github Link:**

https://github.com/rudraaaajoshi/FDSA_BATCH_A/tree/main/Practical-5

Difficulty :

Handling both prev and next pointers was a little difficult.

Error :

The prev pointer of the new head was not set to NULL.

Solution:

Set head->prev = NULL after deleting the first node.

Outcome:

The playlist worked correctly and all operations were performed successfully.

Code(Github) Link:

https://github.com/rudraaaajoshi/FDSA_BATCH_A/blob/main/Practical-5/5.1.cpp

Output:

```
Song A <-> NULL
Total songs = 1
Song A <-> Song B <-> NULL
Total songs = 2
Song A <-> Song C <-> Song B <-> NULL
Total songs = 3
Song C <-> Song B <-> NULL
Total songs = 2

Process returned 0 (0x0)   execution time : 1.037 s
Press any key to continue.
|
```

Key Question:**Problem-1**

When deleting the front node, why does the second node's backward (prev) link create extra responsibility?

Answer:

When deleting the front node, the second node becomes the new head.

Its prev must be changed to NULL.

Otherwise, it will point to the deleted node, causing a dangling pointer.

Practical-4.2**Difficulty:**

Managing circular connections in both lists was a little difficult.

Error :

The last node was not connected back to the first node correctly.

Solution:

Updated the next and prev pointers to maintain the circular connection.

Outcome:

The circular linked lists worked correctly.

Code(Github) Link:

https://github.com/rudraaaaajoshi/FDSA_BATCH_A/blob/main/Practical-5/5.2.cpp

Output:

```
SINGLY CIRCULAR LIST
After joining: 1 2 3
After student 2 leaves: 1 3

DOUBLY CIRCULAR LIST
After joining: 1 2 3
After student 2 leaves: 1 3

Process returned 0 (0x0)    execution time : 1.041 s
Press any key to continue.
|
```

Key Question:**Problem-2:**

Why can circular-list traversal loop forever?

Answer:

A circular linked list does not end with NULL; the last node points to the first.

So traversal should stop when temp == head.

Using temp != NULL would cause an infinite loop.

